

How to run the whole thing

Written in plain words. What sits where, what each file is, what you press, what the second laptop is actually for, and why any of it matters. No prior knowledge assumed.

PART 1 · THE MAP

Two laptops, and what lives on each

Everything below rests on one idea: **the two laptops must not be able to stand in for each other**. If one person could produce both signatures, the second signature would prove nothing at all.

Laptop A – the worker

This machine. It does the actual work.

- The tool itself, at `D:\akhandasih\akhandasih`
- Your signing key
- The record file it produces
- The screen you show people

Laptop B – the witness (the LOQ)

Holds one key and answers questions. Nothing else.

- One small program, `node.py`
- Its own separate signing key
- Its own separate copy of what it signed
- **No case files. None.**

THE MOST COMMON MISUNDERSTANDING

The screen never downloads files from laptop B, because laptop B has no files to give.

That is deliberate. If the witness also stored evidence, anyone who broke into it would get the evidence *and* the key that vouches for it. It holds the key and nothing worth stealing alongside it.

PART 2 · WHEN THE FRONTEND ARRIVES

What to send me, and what I will do with it

1

Put her folder somewhere I can read

Anywhere on the D drive is fine. If she sends a zip, unzip it first. Then tell me the folder name. That is all I need to start.

2

I check whether it runs at all

Some frontends need a build step and some do not. I will find out which, run it, and tell you plainly whether it works before we spend time on anything else.

3

I connect it to the real data

Her pages will be showing invented sample data. I replace that with the real record your tool produced, so what a judge sees on screen is the actual output and not a mock-up.

4

I tell you what is missing, honestly

If something she built cannot be connected because the tool does not produce that information, I will say so rather than invent a number to fill the gap.

WHAT I WILL ASK YOU FOR

Almost certainly nothing. No accounts to create, no keys to buy, no server to rent. The tool writes files, the page reads files. If her version genuinely needs something extra, I will name it and explain why before you spend anything.

The one question I may ask: whether to keep your current screen as the safe fallback. My answer in advance is **yes, keep it** — a demo with no working screen is a lost demo, and having two is free.

PART 3 • RUNNING THE PROTOTYPE**The whole sequence, in order**

Do these in order. Each one depends on the one before it.

1

Put both laptops on the same hotspot

Use **your phone's hotspot, or this laptop's own hotspot**. Do not use campus wi-fi. Many campus networks deliberately stop two laptops from talking to each other, even though both have internet, and you will not discover this until the moment you need it.

2

Start the witness on laptop B, and leave the window open

```
python node.py
```

You should see `Application startup complete`. That window must stay open for the whole demo. Closing it stops the witness.

Closing it does not break anything quietly — the tool will refuse to run and will record the refusal. That is correct behaviour, but not what you want on stage.

3

Check laptop A can reach it

```
curl http://192.168.137.132:8000/witness/pubkey
```

You want a short line of text back containing a `key_id`. Nothing back means either the window in step 2 is closed, or laptop B's firewall is blocking it.

4

Do the actual work on laptop A

```
cd "D:\akhanda SIH\akhanda"  
python scripts/demo_bundle.py --witness http://192.168.137.132:8000
```

About five seconds. This is the step that **creates the files**. It wipes some files, recovers some files, deliberately tries one job with the witness switched off so you can show it refusing, and writes the certificate.

5

Open the screen

```
cd "D:\akhanda SIH\console-app"  
python -m http.server 8080
```

Then open `http://localhost:8080` in a browser. Click **Open chain.json** and choose the file the last step made. Put laptop B's address in the witness box and press **Connect**.

What each file is, in plain words

Everything the tool makes lands in one folder: `D:\akhanda SIH\akhanda\demo_bundle\`

FILE	WHAT IT ACTUALLY IS	USED FOR
chain.json	The record book. Every action, in order, each one locked to the one before it. This is the file you open on the screen.	The main demo
chain_tampered.json	A copy with one entry deliberately altered, to prove the tool catches it.	The tamper demo
certificate.json certificate.txt	The court document. Every line marked with the exact law or standard it answers.	Show a judge
byte_identical_proof.json	For each recovered file, the fingerprint before and after. Identical means the file came back perfectly.	Proving recovery works
recover_result.json	Every file found, and honestly whether any came back incomplete.	Detail if asked
erase_result.json	What was wiped, and whether the wipe was checked afterwards.	Detail if asked
MANIFEST.json	Which steps ran and which failed. A failed step is written down, never hidden.	Honesty record
recovered\	The actual recovered files. You can open them.	Show they really opened

ONE FOLDER YOU MUST NEVER SHARE

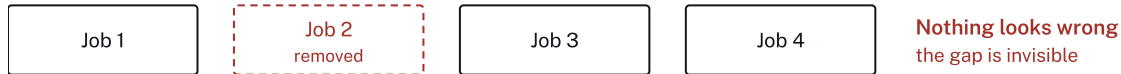
demo_bundle_akhanda_home\ contains **your private signing key**. Never put it on a USB you hand someone, never email it, never upload it. Anyone holding that key can sign records as you.

What gets uploaded, and what never does

THING	GOES ONLINE?	WHY
The screen (the web page)	yes, safe	It only reads and checks. It holds no key.
The record and the certificate	safe to show	They are meant to be shown. They contain fingerprints, never keys.
The tool that wipes disks	never	A disk wiper reachable from the internet is a weapon.
The witness program	never	It holds a signing key. It stays on your own small network.
Your private key	never	It is the one secret in the whole system.

The reasoning, so you can defend it

EVERY OTHER TOOL — each page signed on its own



AKHANDA — each record locked to the one before



The single idea the whole project rests on. Everything else exists to support it.

Why a second laptop

A signature only means something if the person it accuses could not have forged it. Put both keys on one machine and the operator can sign as both. Put the second key on a machine they cannot log into, and the second signature becomes real evidence.

Why it refuses instead of carrying on

If a missing witness simply meant a weaker record, anyone could unplug it and quietly do unwatched work. Because the tool *stops* and *writes down that it stopped*, unplugging the witness becomes a way to leave a mark, not a way to avoid one.

Why the check happens first

Wiping cannot be undone. Checking the witness afterwards would be asking permission for something already done. The check happens before a single byte is destroyed.

Why the browser does the checking

If our own program says "this is valid", you are trusting us. When the page recalculates it in front of you, on your own machine, you are trusting arithmetic. A doubter can open the browser tools and watch it happen.

How this answers SIH26149

The statement asks for an *integrated secure data erasure and advanced file recovery tool for digital forensics and data sanitization*. Taking each word seriously:

WHAT IS ASKED	WHAT WE DO	STATE
Secure data erasure	Wipes whole disks and individual files, checks the wipe afterwards, and records the honest grade achieved	built
Advanced file recovery	Recovers deleted files, including files broken into pieces, and returns them byte for byte or not at all	built
Integrated	Both live in one record book. This is the part nobody else does — elsewhere wiping and recovery are separate products with separate logs	built
For digital forensics	Produces the two-person certificate Indian courts have required since July 2024	built
Data sanitization	Follows the international NIST standard and states which grade was truly reached	built
Tamper-evident reporting	Remove a record and verification fails, naming the exact one	proven
Highest wipe grade on solid state drives	The code is written and tested, but needs lab hardware we could not borrow	not tested

THE SENTENCE THAT WINS THE ROOM

"Every tool in this space signs each certificate on its own. That proves one page was not edited. It does not prove the file is complete. **We are the only ones who can show you which record is missing.**"

PART 7 • GROWTH

Where it goes from here

How it grows without getting expensive

- **One witness serves a whole laboratory.** Twenty workstations can all send their records to the same second machine. The cost does not rise for each new user, which is exactly how the tools we compete with make their money.
- **The record stays fast as it grows.** Tested on five hundred records, a tampered one is found in thirty-seven thousandths of a second. Case files can build up for years.

- **Nothing to install centrally.** No server room, no database, no yearly renewal. A laboratory that adopts it is not signing up to a dependency.

The upgrades that would matter most

UPGRADE	WHAT IT WOULD CHANGE	EFFORT
The presence device	Lifts us from the second highest trust level to the highest, by proving a human was physically present, not just a running program	designed already
Government certification	STQC approval is what lets a real laboratory buy it. This is the single biggest gap between a prototype and a product	months, and money
One real laboratory pilot	Turns "it works on our laptops" into "it works on their evidence". Worth more than any feature	a relationship, not code
Outside time-stamping	Today the clock is your own computer's. An independent time source closes the last gap a lawyer could pick at	partly built
Highest-grade wiping on solid state drives	Written and tested, but unproven until we borrow real drives	needs hardware only

THE HONEST LIMIT, AND WHY SAYING IT IS A STRENGTH

The record proves nothing was **changed**. It cannot prove nothing was **hidden** before it was ever written down. No ledger in the world can. Recording refusals narrows the gap; nothing closes it.

Say this before a judge asks. A team that names its own limit is believed on everything else. A team caught hiding one is believed on nothing.

PART 8 • BEFORE YOU PRESENT

The list to run through

CHECK	HOW YOU KNOW IT IS FINE
Both laptops on your own hotspot	Not campus wi-fi. Not negotiable.
Witness running on laptop B	The window says startup complete, and stays open
Laptop A can reach it	The curl command returns a key id
Fresh records made <i>with</i> the witness	The screen shows a co-signer on the first row
The screen loads and turns green	Integrity reads verified

CHECK	HOW YOU KNOW IT IS FINE
Connect shows the match line	The live key equals the one that signed the record
You have rehearsed tamper and restore	Once, out loud, before the day
One power strip	The venue gives one socket per team

MOVE THE WITNESS OFF THE USB STICK

You ran `node .py` from the USB drive. If that stick is unplugged, or is needed for something else, the witness stops. **Copy the folder onto laptop B's own hard drive and run it from there** before the day.

Akhanda · अखण्ड, "unbroken" · Syntax Squad · GITAM (Deemed to be University), Hyderabad · Smart India Hackathon 2026, problem statement SIH26149, sponsored by NTRO.

Every figure here was measured on the machines described, on 3 September 2026.